Faculty of Computing

Universiti Teknologi Malaysia

Data Structure & Algorithm

Lab Exercise 01

Instruction

- **Write down** all answers for each question in Part 1-3. Upload the answers (scanned photo) in E-learning Forum by replying to the main thread with your name, matric no and perform submission of the source code (.cpp) for Part 4.

Part 1:

1. Given the following program, show the complete recursive process for the answer find (1,5).

```
1   int find(int a,int b) {
2   if (a==0)
3   {
4   return (b+1);
5   }
6   if(b==0)
7   {
8   return(find(a-1,1));
9   }
10  else
11  {
12  return (find(a-1,find(a,b-1)));
13  }
14 }
```

Write down the trace using:

```
int find(1,5) {
if (___==0)
{
return (___+1);
}
if(___==0)
{
return(find(___-1,1));
}
else
{
return (find(___-1,find(___,___-1)));
}
}
```

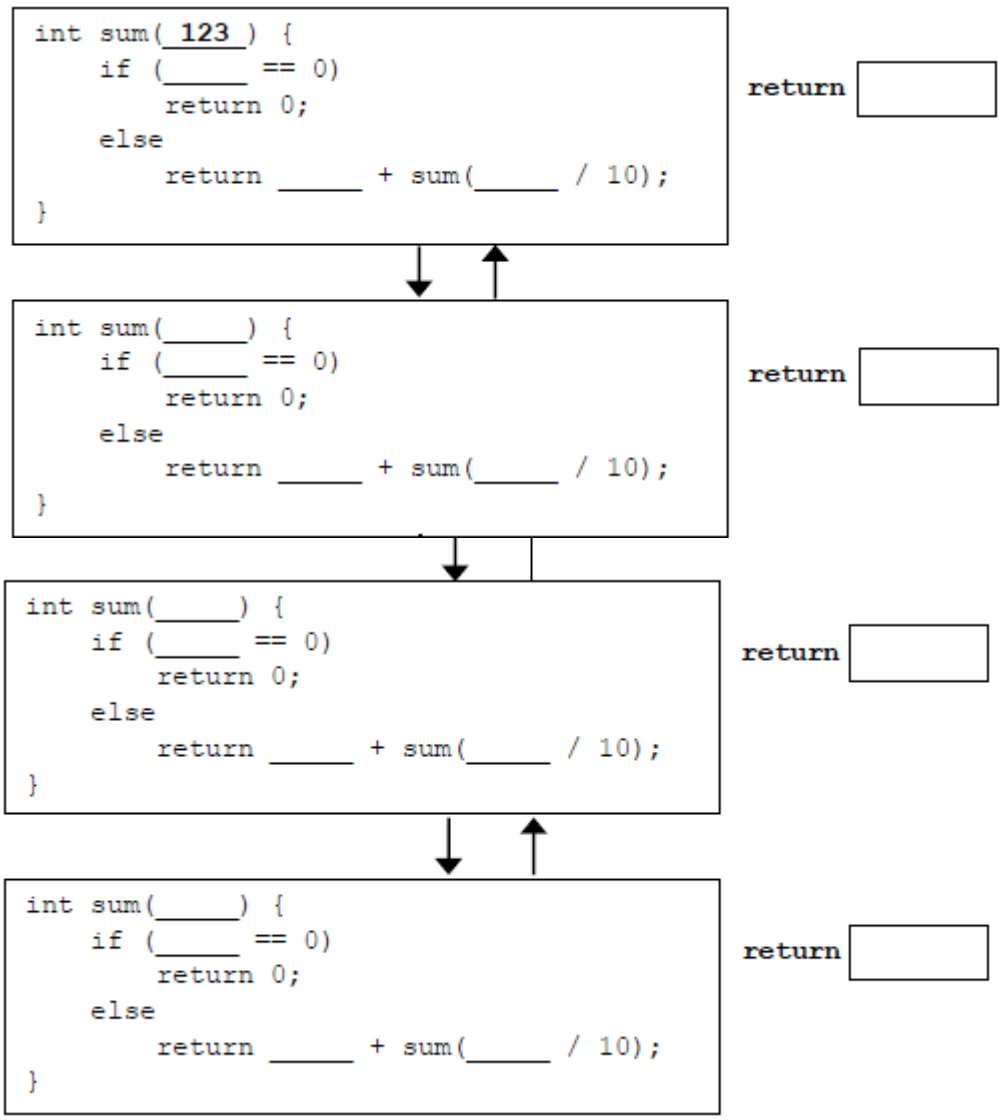
```
return ___
```

Part 2:

1. Given the following program, Trace the recursive calls of the sum function in the above program. Write your answer by completing the tracing template below when n = 123.

```
1   int sum(int n) {
2   if (n == 0)
3   return 0;
4   else
5   return n + sum(n / 10);
6   }
7   int main() {
8   cout << sum(123);
9   return 0;
10  }
```

Fill in the underlined blank spaces/boxes with values ONLY where applicable



Part 3:

1. Given the following program:

```
1   #include <iostream>
2   using namespace std;
3
4   void reverseString(string str, int index) {
5       if (index < 0)
6           return;
7       cout << index << ":" << str[index] << "\n";
8       reverseString(str, index - 1);
9   }
10
11  int main() {
12      string text = "ALGORITHM";
13      reverseString(text, text.length() - 1);
14      return 0;
15  }
```

a. Trace the recursive algorithm and write the output.

b. What is the base/terminal case of the `reverseString` function

c. What is the general case of the `reverseString` function

Part 4:

1. Write a recursive function named power(int base, int exponent) that calculates and returns the value of base raised to the power of exponent.The function should stop the recursion when the exponent becomes zero. The results for input with base = 2 & exponent = 4 suppose to give output of 16. Submit the power.cpp file in the submission link in Elearning.

Use the following recursive relationship:

$$power(base, exponent) = base \times power(base, exponent - 1)$$

and

$$power(base, 0) = 1$$

Input: base = 2,
exponent = 4
Output: 2^4 = 16